HS4002: Lecture 10B

Dimensionality Reduction II

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Introduction to Multiple Correspondence Analysis

Multiple Correspondence Analysis

MCA is a dimensionality reduction technique that we use to examine relationships among categorical variables. It's most often used in descriptive analysis that looks at the relationality among a group of categorical variables.

Why MCA?

MCA is a somewhat technique in the social sciences and data science. Most famously, it's associated with the work of Pierre Bourdieu.

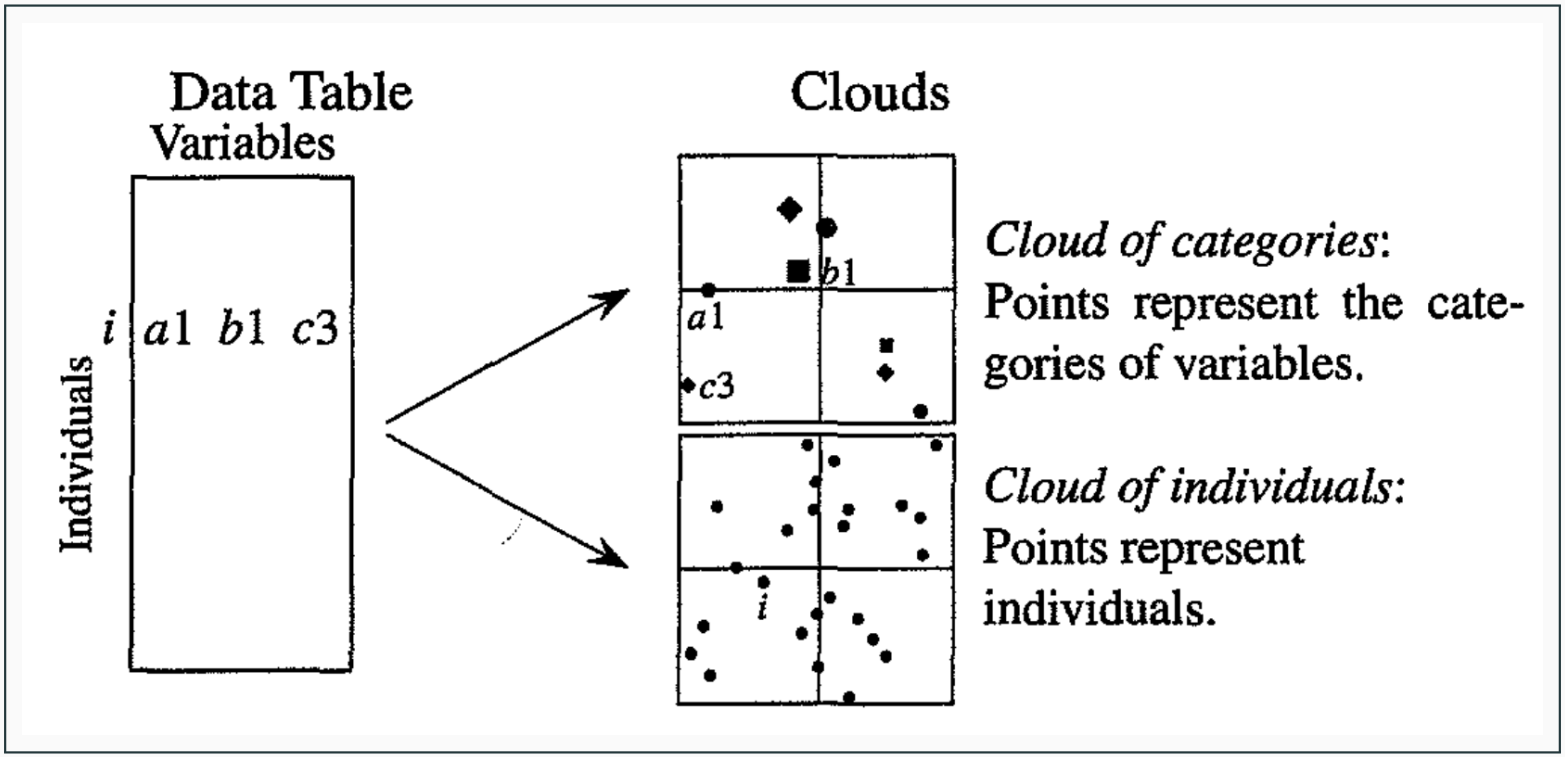
1. Like PCA, we can use it to reduce the dimensionality of our data.
2. In addition, we can use MCA to produce a projection of the categories themselves, something we don't do with PCA.

Intuition behind MCA

Imagine the data as a matrix (math for “table") where each row is a individual, and each column is - some - categorical variable (e.g. gender, binary indicators for likes/dislikes).

MCA is a way of representing this space/table using fewer dimensions than before, while closely reproducing the similarities among the rows and columns of the data table, so that proximity in the space indicates similarity of categories and of individuals.

Intuition behind MCA



Some Differences between MCA and PCA

- MCA requires categorical variables.
- Instead of explaining variance (PCA), we are explaining inertia (MCA).
- MCA produces a "quantification" of the similarities and differences among the different categories.

Math Behind MCA

Math Behind MCA

There are many ways, but usually through the spectral decomposition of the Burt Matrix.

We start with a table

Table 1.1 *Taste Example*. Extract from the Individuals × Questions table.

	TV	*Film*	*Art*	*Eat out*
1	Soap	Action	Landscape	SteakHouse
7	News	Action	Landscape	IndianRest
31	Soap	Romance	Portrait	Fish&Chips
235	News	Costume Drama	Renaissance	FrenchRest
679	Comedy	Horror	Modern	Indian
1215	Soap	Documentary	Landscape	SteakHouse

Note: A row corresponds to the *response pattern* of an individual. For instance, the pattern of individual #235 is (*TV*–News, *Film*–Costume Drama, *Art*–Renaissance, *Eat out*–FrenchRest).

Math Behind MCA

Suppose there are Q categorical variables, and any variable q has J_q categories. Let Z be an indicator matrix of size $(n \times J)$, such that $J = \sum_{q=1}^{Q} J_q = J_1 + ... + J_Q$ and $n$is equal to the total number of observations.

What such an indicator matrix looks like

Qu. 1				*Qu. 2*				*Qu. 3*				*Qu. 4*			
W	*w*	*H*	*?*	*W*	*w*	*H*	*?*	*W*	*w*	*H*	*?*	*W*	*w*	*H*	*?*
1	0	0	0	0	0	1	0	0	1	0	0	0	1	0	0
0	1	0	0	0	0	1	0	0	0	1	0	0	1	0	0
0	0	0	1	0	0	1	0	0	0	1	0	0	1	0	0
0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1
0	0	0	1	0	0	0	1	0	0	0	1	0	0	0	1
1	0	0	0	0	0	1	0	0	1	0	0	1	0	0	0
⋮				⋮				⋮				⋮			

Math Behind MCA

Then the Burt matrix is the symmetric partitioned matrix

$$B = Z^T Z.$$

Perform singular value decomposition on the Burt Matrix to obtain the MCA solution containing the principal inertias.

... eventually a biplot!

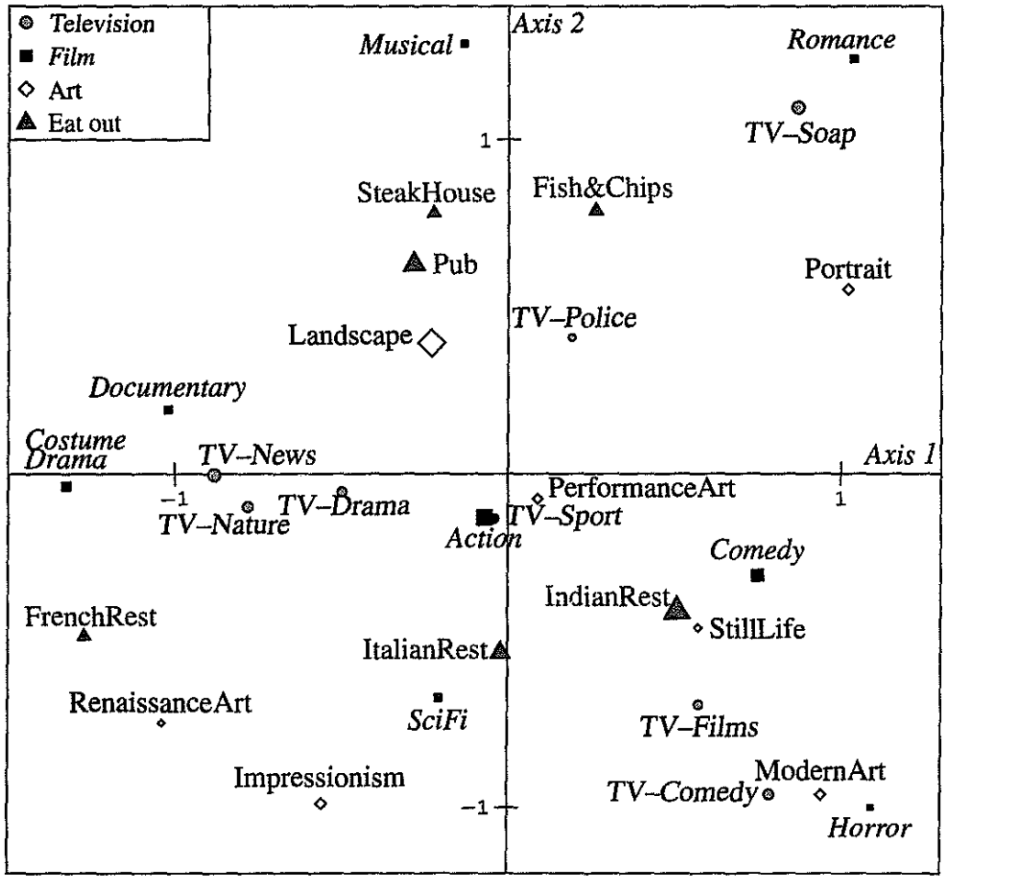


Figure 1.2 *Taste Example.* Cloud of the 29 categories of the four questions in Principal plane 1-2.

Applied MCA

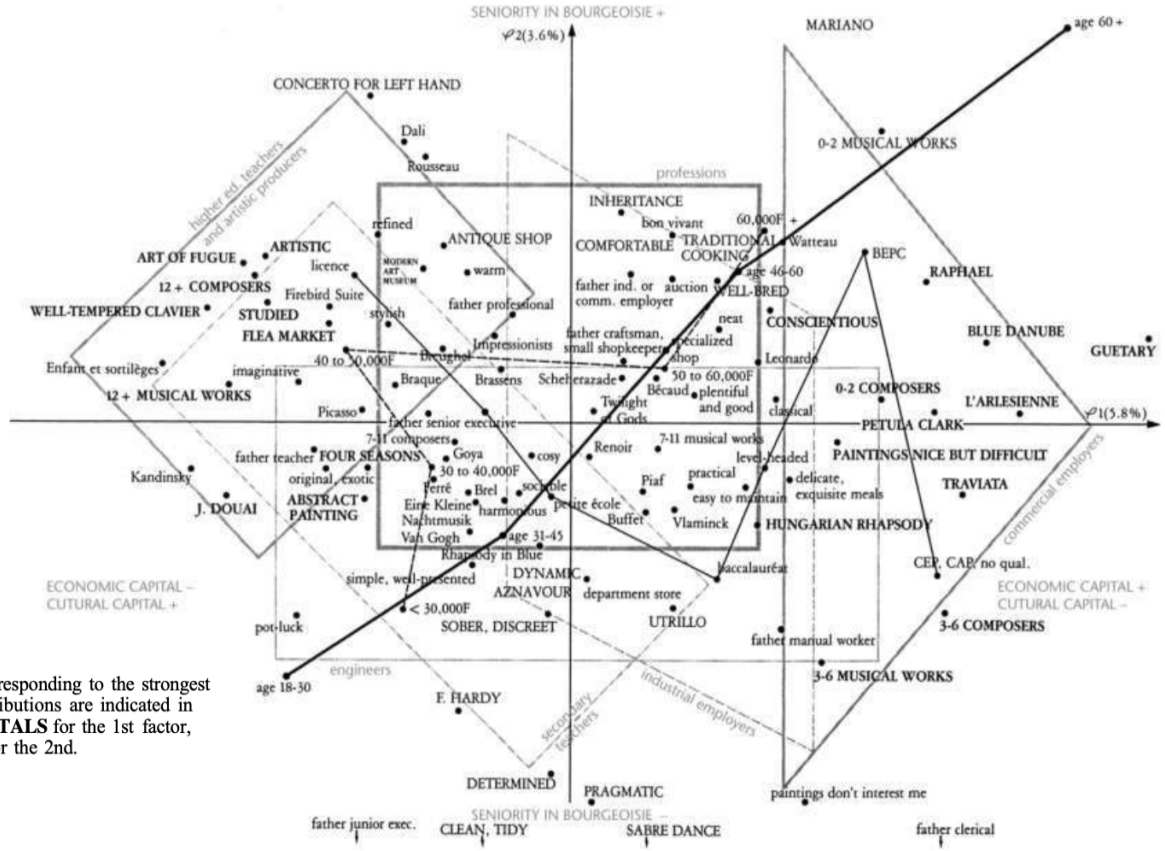
Steps behind MCA

1. Choose variables of interest (make sure they are categorical)
2. Conduct MCA.
3. Examine contributions to each dimension.
4. Plot biplot.
5. Make appropriate interpretations.

Biplots

- Biplots lets you project the two most important dimensions into a 2D space.
- The space summarizes the relationships among the categorical variables within the dimension.
- The meaning of any given category must understood in relation to the positions of the other categories.
- The closer two categories are, the more similar they are.
- The further two categories are, the more dissimilar.
- MCA tries to produce a two-dimensional project that best preserves the relationships among all of the different categories and persons in the data.

How Bourdieu used it



The items corresponding to the strongest absolute contributions are indicated in **BOLD CAPITALS** for the 1st factor, CAPITALS for the 2nd.

Figure 11 (shown in black) Variants of the dominant taste. Analysis of correspondences. Plane diagram of 1st and 2nd axes of inertia: the space of properties.

Figure 12 (shown in grey) Variants of the dominant taste. Analysis of correspondences. Plane diagram of 1st and 2nd axes of inertia: the space of individuals of the different fractions.

What did Bourdieu see?

- "Thus on the left side ... are those who, with the lowest incomes, have the greatest competence, who know the largest number of musical works and composers, who prefer the works demanding the 'purest' aesthetic disposition ..."
- "On the right side are those who receive the highest incomes and have the lowest competence, who know few musical works or composers ..."

Example: Childress, Rawlings & Moeran's (2017) work on the Booker Prize.

Childress, Rawlings & Moeran (2017)

Which positions do publishers occupy when they submit a novel to the Booker Prize, and which of those positions does the jury consecrate?

Childress, Rawlings & Moeran (2017)

- The rows are novels submitted to the Booker Prize, 1983–1991.
- A publisher may submit two novels a year. These are the books the house itself put forward.
- We have info on the following:
  - → the publisher, the author, and the text
  - → Place, gender, and race of the author and of the main character
  - → where the story is set
  - → genre and length
  - → whether the house is old, in Bloomsbury, a conglomerate, or part of the Cape Group.

The MCA

- The categories are nominal, so they use an MCA.
- The map is the space of nominations: the positions the publishers collectively take up.

The MCA

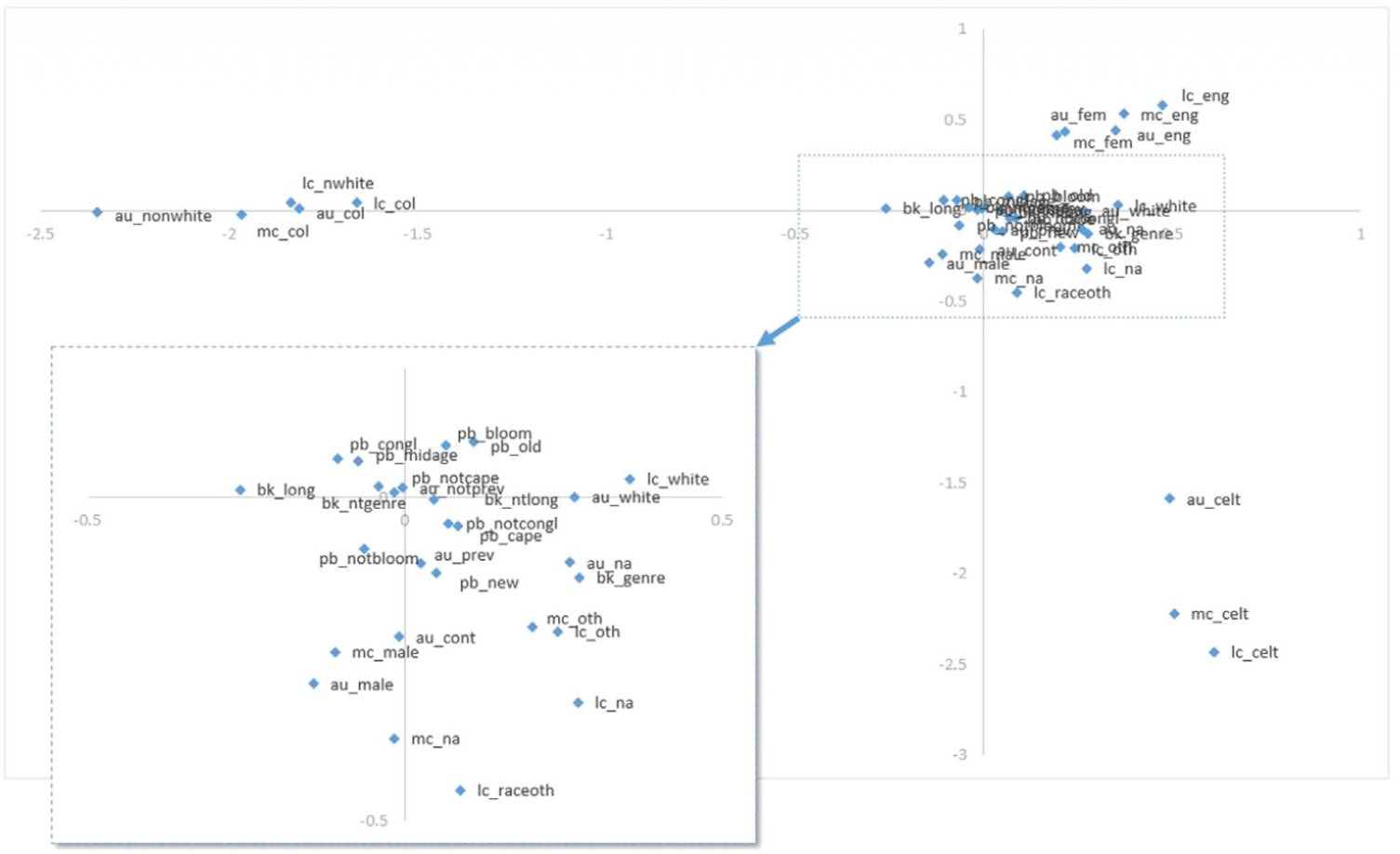


Fig. 1. Multiple Correspondence Analysis Results.

The insets enlarge the clump at the centre.

Reading the map

- On the first dimension, English authors, English settings, English main characters, and white authors sit together. Authors and characters from the former colonies sit at the other pole. Female authors and female characters sit with England. Male authors and male characters sit toward the colonies.
- On the second dimension, the Celtic countries sit opposite England. North American novels sit between the two.
- An author's place lands on the story's place, and an author's gender lands on the main character's gender. The publishers mostly clump near the centre.

From the map to a model

They use MCA dimensions as independent variables in a linear model.

Results

Table 3
Results from Multinomial Logistic Regression of Likelihood of Prize Shortlisting and Winning (Nominated Omitted Category).

	Shortlisted vs. Nominated	Won vs. Nominated
F1: English vs. former colonial authors and stories	−0.087	−2.097*
	(0.468)	(1.022)
F2: English vs. Celtic authors and stories	−0.955*	−0.827
	(0.488)	(1.742)
F3: Continental European authors and stories	0.751	−3.801
	(0.651)	(3.078)
F4: North American authors and stories	1.054	0.726
	(0.610)	(2.404)
F5: Celebrated authors writing "literature" for high-status English publishers	1.843***	2.506*
	(0.424)	(1.047)
F6: Female authors and characters	−0.956	−4.436*
	(0.560)	(2.243)
F7: Historic publishers	−0.401	−1.7
	(0.449)	(1.147)
F8:Conglomerates	1.452**	−1.452
	(0.545)	(1.806)
F9: Non-North American conglomerates	−0.687	0.183
	(0.445)	(1.314)
Constant	−4.088***	−5.054***
	(0.677)	(1.371)
Log-likelihood	−204.382	
Number of observations	1094	
Degrees of freedom	34	
"		

Note: Model includes t-1 indicator variables for years.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

Thanks for listening :)